MEng Rubric – Project Planning Report

Project Overview – 5%

Level	Criteria
0	Section missing, or unable to meet the next rubric level.
Unsatisfactory	Overview provides a very limited understanding of the project with little to no added value from the supervisor-provided project description. Project diagram may be very high-level, unclear, or missing entirely.
Pass	Student provides an overview of the main project aim that clearly and concisely sets out what the project must achieve and the importance of the work. A diagram is included that shows consideration of the main system components (hardware and/or software) or processes and how these are interconnected.

Pass requirement must be met before any elements below can be included.

++	Project overview makes use of **suitable** references to support the **engineering context** of the project (e.g. use of academic papers for key concepts that will be applied).			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Project overview makes use of **suitable** references to support the **commercial context** of the project (e.g. using market data to show consumer demand).			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	The project diagram makes good use of **stylistic features** (colour, font styles, groupings, shapes, etc) to intuitively **separate sub-systems and components** within the overall project. Use of styles are **consistent**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Project diagram demonstrates complete understanding of **critical component level design requirements** (for example a MCU block might include peripheral/memory/clock needs, software routines may include API, algorithm, limitations, data type, execution time requirements).			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	The project diagram **clearly shows**, and where necessary **explains**, **dependency** between parts.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

Originality – 25%

Level	Criteria
0%	Section missing, or unable to meet the next rubric level.
Unsatisfactory	Student demonstrates very limited background reading and is missing a direct explanation of the originality of the work or the explanation of this is poor and not well supported by their referencing.
Pass	Student provides a suitable explanation of the originality of the project, highlighting where it fits within commercial/scientific context along with differentiating from existing intellectual property. Student provides a suitable explanation of the originality of the project and attempts to demonstrate the need for this through assessment of the current commercial/scientific context.

Pass requirement must be met before any elements below can be included.

+++	The need for the project is clearly identified through assessment of **both** the wider **background** and a **comparative** look at **existing IP/current practice**. There is **clear differentiation** from existing IP/current practice.

Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

+++	Student acknowledges the **limitations** their project will have when **compared** to other work (e.g. less comprehensive testing facilities, only prototype manufacturing equipment). These are both discussed and **justified** with relation to originality. Student can **quantify** and **extrapolate** the **impact** of the limitations to their project **specification**.

Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Student explains project originality not only in the final deliverable, **but also in the development approach**.

Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	**Multifaceted** novelty combining two or more of: new technique/approach/algorithm, interdisciplinary approach, translation from one technology area to others, low-cost solution with comparable performance to current practice.

Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

Specification – 7.5%

Level	Criteria
0	Section missing, or unable to meet the next rubric level.
Unsatisfactory	Provided specification is very limited and shows little to no expansion on the original project description. Specification points given may be poorly defined or unclear.
Pass	Student provides a list of relevant requirements for the final project deliverable that can be used to measure the success of the project. Specification points are suitably prioritised with additional requirements over the main aim given as "stretch goals" and grouped in a suitable way for the project (for example splitting hardware and software requirements).

Pass requirement must be met before any elements below can be included.

++	Specification points are all **quantitatively justified** with respect to the project aim(s) and **wider literature** (e.g. H&S, legal requirements, engineering standards, best practice, etc). With inclusion of **suitable references**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	There is clear **assessment** of the expected **challenge** posed by each specification point.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Each requirement includes a succinctly clear **explanation** of how it will be **tested** including the **criteria for success**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Specifications are shown to be both **attainable** (with the facilities available to the student) and **realistic** (in terms of the time frame of the project).			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Clear ambition to achieve specification **beyond** commercial-off-the-shelf or state-of-the-art research-grade system. There is evidence that such ambition is **realisable**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

Methodology – 7.5%

Level	Criteria
0	Section missing, or unable to meet the next rubric level.
Unsatisfactory	Student attempts to provide a development process for the project, with a limited inclusion of key tools and components, without consideration of the limitations related to these.
Pass	Student provides a suitable development process for the project, highlighting some key tools and components that will be used within each step, and outlining the obvious limitations related to these. Selected tools are suitably justified with respect to the project aim/specification/resources/time/etc.

Pass requirement must be met before any elements below can be included.

++	Discussion of the **implications** of **inter-dependencies** between different steps.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

+++	Each step of the methodology is **assessed** in terms of the **challenge** posed to the student based on their individual experience and skill set. **Additional tasks** related to these are explained (e.g. use of a piece of software may require additional tasks to become familiar).			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

+++	**Limitations** are explained and assessed with **relation** to the project development (e.g. using a licenced software may incur cost but be beneficial if the student already has experience).			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Student effectively **combines/synthesizes/translates** existing methodologies leading to a **bespoke** solution which meets the specifications and **accounts** for all limitations.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

Risk Management and Mitigation – 15%

Level	Criteria
0	Section missing, or unable to meet the next rubric level. Risks presented are not related to project management.
Unsatisfactory	Risks presented may be missing mitigation actions. Risks presented may not be relevant to the project or very generic with little explanation.
Pass	Student provides a list of some obvious expected risks that could impact completion of the project with inclusion of risks that are specific to the project methodology. Risks are accompanied by suitable mitigation actions.

Pass requirement must be met before any elements below can be included.

++	The potential **impact** of risks on the project are **realistically quantified** in terms of time/cost/change in spec. Risks are **sorted** by impact.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Mitigation actions are **realistic** given the **resources** available to the project.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	The **likelihood** of risks occurring are **quantified** with a **method applied** to sort the **importance** of each risk.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	There is clear indication of **when** each risk is likely to occur in the project and this is **considered** as part of the mitigation actions.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	All critical risks to the project are **fully mitigated** and **quantified (rated) pre and post** mitigation.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

Time Plan – 15%

Level	Criteria
0	Section missing, or unable to meet the next rubric level.
Unsatisfactory	Time plan is very basic, may be missing Gantt chart and just given as a list of tasks. Tasks show little thought and either reproduce information from the supervisor project handout or are a repeat of the specification points.
Pass	Student provides a Gantt chart that outlines the expected progress of the project, broken down into major tasks with completion points of deliverables and milestones indicated on chart. Gantt chart includes stages at which progress reviews will be carried out with an explanation is given for where the project is expected to be at each review.

Pass requirement must be met before any elements below can be included.

+	**Dependency** between tasks is clearly indicated and **explained**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

+	Clear **link** of Gantt chart tasks to project **specification** and **methodology**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Gantt chart and explanation demonstrates **consideration** of **risks** and mitigations with use of **parallelisation**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Gantt chart has a **visible cycle** of develop, test, document.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	**Realistic** time planning and **work loading** (based on # hrs per week).			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Gantt chart **demonstrates** points of **uncertainty** (e.g. via error bars) and the **impact** on following tasks is **explained**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

Interview – 25%

Provide slide template with main headings (title page, block diagram, and time plan).

Level	Criteria
0	Student did not attend interview. (Defence)
Unsatisfactory	Explanation of project is limited and confused. No coherent plan and/or risks presented.
Pass	Student is able to explain their project aim and expected outcome. The time plan for the project and potential risks are explained to the moderator. When questioned about risks to the project, student is able to demonstrate/discuss beginning steps that have been taken to mitigate these.

Pass requirement must be met before any elements below can be included.

++	Project aim is clearly explained, and when **questioned** student shows a good understanding of how the aim **will be met**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Student **convincingly** shows that the project can be **completed** with the **resources available to them**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Student reflects on the **personal limitations** of their planning process and their **ability** to complete the project. Questioning shows that limitations are **convincingly managed**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Through questioning and presentation, the student can **convincingly justify** the project **undertaking**.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)

++	Given presentation (both **provided materials** and students **delivery**) is **effective** in its explanation of the project and plan.			
Not present (0%)	Some (40%)	Most (60%)	Fully met (80%)	Exceeded (100%)